Final Project

Erosh Boralugodage

May 10, 2023

Abstract

Identifying a person in a video is challenging, especially when the same person appears in multiple videos and times. In fields like entertainment and law enforcement, it is necessary to accurately and efficiently identify the person in videos. Developing a machine learning model that can identify the same person in multiple videos can be useful in these industries.

Contents

1 Project **1**

2 Data set **2**

3 Data preprocessing **2**

4 model develpment **3**

5 Model evaluate **4**

6 Model Application **6**

7 conclusion **7**

1 Project

The objectives of this project are to develop a Convolutional Neural Network (CNN) model that can accurately identify a person from a video and to use the model to identify the same character in other videos.

TASK 1

Select and extract frames from the videos and Data Preprocessing

2 Data set

Videos downloaded from YouTube in different angles, resolutions, and lighting conditions. Collect a diverse dataset of videos that contain the target character. Videos I have selected with different speed, background and lighting. Exclusive One-on-one with President Biden on 2024 re-election bid, debt limit and more.mp4 ENGLISH SPEECH BARACK OBAMA The People of India (English Subtitles).mp4 President Barack Obama delivers stirring speech in Parliament.mp4 Vice President Harris Talks Biden's Marijuana Pardon, 2022 Midterms and Abortion Rights.mp4

3 Data preprocessing

select frames carefully to identify the character manually and make a daabase chracter and non character. Then upload and create training, validation and test data sets ratio 0.7:0.15:0.15 . Training data set contain 1050images with both classes and 224 images in validation data set.

```
[ ] # Set the paths to the original data and the destination folders for the subsets
    original_data_dir = '/content/dataset'
    train_dir = '/content/dataset/train'
    validation_dir = '/content/dataset/validation'
    test_dir = '/content/dataset/test'

    # Set the percentage of data to use for each subset
    train_percentage = 0.7
    validation_percentage = 0.15
    test_percentage = 0.15
```

Figure 1: dataset

TASK 2

Develop a CNN model that can accurately identify the target character from the images extracted from the videos.

4 model develpment

The model should be trained using the collected dataset of images. Model reach 100

```
from tensorflow import keras
import tensorflow as tf
from tensorflow.keras import layers

#define data augmentation pipeline

data_augmentation = keras.Sequential(
  [
    layers.RandomFlip("vertical"),
    layers.RandomCrop(256,256),
    layers.RandomContrast(0.2),
    layers.RandomRotation(0.1),
    layers.RandomTranslation(0.1,0.1),
    layers.RandomZoom(0.2),
  ]
)
```

```
[ ] from tensorflow.keras.models import Sequential
    from tensorflow.keras.layers import Dense, Conv2D, MaxPooling2D, Flatten

    #setup CNN archtecture
    model = Sequential([
        data_augmentation,
        Conv2D(32, (3,3), activation ='relu', input_shape = input_shape),
        MaxPooling2D((2,2)),
        Conv2D(64, (3,3), activation ='relu', input_shape = input_shape),
        MaxPooling2D((2,2)),
        Conv2D(128, (3,3), activation ='relu', input_shape = input_shape),
        MaxPooling2D((2,2)),
        Flatten(),
        Dense(128, activation = 'relu'),
        Dense(1,activation='sigmoid')

    ])
```

Figure 2: training accuracy vs validation accuracy

TASK 3

evaluate the model with validation and test data set.

5 Model evaluate

Evaluate the model on test data and model reached 98

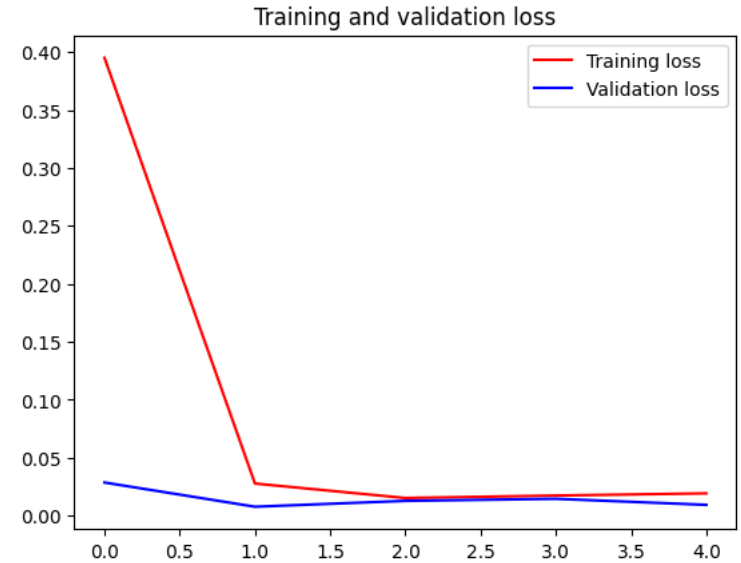
```
#Evaluate the model on test data
test_datagen = ImageDataGenerator(rescale=1./255)

test_generator = test_datagen.flow_from_directory(
    test_dir,
    target_size=input_shape[:2],
    batch_size = batch_size,
    class_mode = 'binary'
)

model.evaluate(test_generator)
```

```
Found 227 images belonging to 2 classes.
8/8 [==============================] - 1s 147ms/step - loss: 0.0562 - accuracy: 0.9868
[0.05621030926704407, 0.9867841601371765]
```

Figure 3:




```
<Figure size 640x480 with 0 Axes>
```

Figure 4:

TASK 4

6 Model Application

Use the trained model to identify the same character in other videos. The videos should be pre-processed to extract frames and create a dataset of images. The model should be applied to the dataset of images to identify the character. The model used to identfy the character on a different video to identify the character are output starting time/frame no and end time/frame the character identified. www.youtube.com/watch?v=k0jJL$_Y FyIUt = 42s$

```
212/212 [==============================] - 5s 24ms/step
character appears from frame 269 to 277.
character appears from frame 1531 to 1628.
character appears from frame 1903 to 1911.
character appears from frame 2183 to 2191.
character appears from frame 2718 to 2726.
character appears from frame 2925 to 3001.
character appears from frame 3166 to 3174.
character appears from frame 3470 to 3646.
character appears from frame 3737 to 3907.
character appears from frame 3982 to 4298.
character appears from frame 4336 to 4481.
character appears from frame 5092 to 5100.
character appears from frame 5499 to 5507.
character appears from frame 6446 to 6454.
character appears from frame 6709 to 6716.
Character appears from8.97s to 9.27s
Character appears from51.03s to 54.30s
Character appears from63.43s to 63.73s
Character appears from72.77s to 73.07s
Character appears from90.60s to 90.90s
Character appears from97.50s to 100.07s
Character appears from105.53s to 105.83s
Character appears from115.67s to 121.57s
Character appears from124.57s to 130.27s
Character appears from132.73s to 143.30s
Character appears from144.53s to 149.40s
Character appears from169.73s to 170.03s
Character appears from183.30s to 183.60s
Character appears from214.87s to 215.17s
Character appears from223.63s to 223.90s
```

Figure 5: Final results

7 conclusion

The model worked fine on test data set and video. I believe if there more data frames on the data set it will enhance the accuracy. This project aims to develop a CNN model that can accurately identify a character from a video and use the model to identify the same character in other videos. The software tool developed from this will be beneficial to industries such as entertainment and law enforcement, which require accurate and efficient identification of characters in videos. The project will be carried out in a step-by-step manner, starting with data collection and preprocessing, followed by model development and evaluation, and concluding with the deployment of the software tool.